Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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Abstract

Large pre-trained language models store factual knowledge in their parameters, but their ability to access and manipulate knowledge is limited, and they often hallucinate facts on knowledge-intensive tasks. We explore a general-purpose fine-tuning recipe for Retrieval-Augmented Generation (RAG). We build RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever (DPR). We compare two formulations: RAG-Sequence, which uses the same retrieved document across the whole sequence, and RAG-Token, which can select different documents per generated token. RAG models achieve state-of-the-art results on open-domain question answering benchmarks, generating more factual, specific, and diverse responses than parametric-only seq2seq baselines.

1. Introduction

Pre-trained neural models like BART and T5 cannot easily expand or revise their memory, can hallucinate, and cannot produce verifiable citations. Our hybrid RAG model couples parametric sequence generation with a non-parametric retriever accessing 21 million 100-word passages from Wikipedia.

2. Architecture and Training
The retriever uses Dense Passage Retrieval (DPR) with bi-encoder architectures. For a query x and passage z, the retrieval probability p(z|x) is proportional to exp(q(x)^T d(z)). The generator uses BART-large with 400M parameters. The model is trained end-to-end without direct supervision on which document to retrieve.

3. Experimental Results
On Natural Questions, RAG achieved 44.5 Exact Match (EM), outperforming the closed-book T5-11B baseline (34.5 EM) while using less than 10% of the parameters. On Jeopardy question generation, human evaluators rated RAG responses as 68% more factual than standard BART. 4. Limitations
RAG depends heavily on retriever recall; if the top-k passages do not contain the answer, the generator cannot recover the true entity.